ANKIT SHUKLA

Staff / Lead Systems Software Engineer | C++ | Linux | Distributed Systems | GPU & Graphics | Embedded

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PROFESSIONAL SUMMARY

Systems Software Engineer with **14+ years of experience** designing and developing C/C++ software across Linux, QNX, Windows, and embedded platforms. Strong expertise in **Modern C++, Linux systems programming, multithreading, IPC, networking, GPU/graphics stacks, device-driver-adjacent development, and performance/debugging**.

Recent experience building **C++/Go/Python control-plane microservices** for resource-management workloads using **gRPC, REST, Kubernetes, SQL and NoSQL databases**. Earlier experience includes GPU driver and OpenGL/LLVM graphics development, automotive embedded systems, and safety-critical avionics software.

Experienced technical lead who has **led and mentored a team of six engineers**, owned features from architecture and design through implementation and testing, and supported production systems through debugging and root-cause analysis.

CORE EXPERTISE

- **C/C++:** Modern C++11/14/17, STL, Templates, RAII, Smart Pointers, Move Semantics, Concurrency
- **Linux & Systems:** Linux, POSIX, IPC, TCP/IP, Sockets, Multithreading, Synchronization, Memory Management
- **Distributed Systems:** Microservices, gRPC, REST APIs, Resource Management, SQL/NoSQL
- **Cloud & Containers:** Kubernetes, Docker, AWS, Azure, Cloud-Native Services
- **GPU & Graphics:** GPU Drivers, OpenGL, DRM, LLVM Graphics Stack, Hardware-Accelerated Rendering
- **Embedded:** Embedded Linux, QNX, WinCE, LynxOS, ARM, x86, Device Drivers
- **Automotive:** CAN, Infotainment, Instrument Clusters
- **Safety-Critical:** DO-178B, ARINC 429, ARINC 664/AFDX
- **Engineering:** System Design, Debugging, Root-Cause Analysis, Code Reviews, Technical Leadership

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TECHNICAL SKILLS

Languages: C, C++, Python, Go, Java

C++: C++11/14/17, STL, Templates, Smart Pointers, RAII, Move Semantics, Lambda Expressions, Design Patterns, SOLID

Systems & Linux: Linux, POSIX, IPC, Socket Programming, TCP/IP, Multithreading, Synchronization, Memory Management, Performance Optimization

Distributed Systems & APIs: Microservices, gRPC, REST, JSON/XML, Resource Management

Databases: SQL, NoSQL, Data Modeling

GPU & Graphics: OpenGL, GPU Driver Development, DRM, LLVM-based Graphics Pipeline, Hardware-Accelerated Rendering, Radeon GPU Architecture

Embedded & RTOS: Embedded Linux, QNX, WinCE, LynxOS, ARM, x86, Kernel-adjacent Development, Device Drivers

Cloud & Containers: Kubernetes, Docker, AWS, Azure

Testing: GoogleTest, GoogleMock, VectorCAST, Ginkgo, TestRail

Build & CI/CD: CMake, Make, Bazel, Maven, MSBuild, Jenkins

Debugging: GDB, Trace32/Lauterbach, Core Dump Analysis, Root-Cause Analysis

Security & Compliance: OpenSSL, SBOM, JFrog Artifactory, Black Duck

Tools: Git, Gerrit, Bitbucket, Jira, IBM Rational DOORS

PROFESSIONAL EXPERIENCE

Cohesity Inc. — Software Engineer 4, Resource Manager (Heimdall)

Dec 2024 – Present | Pune, India

- Designed and implemented core components of **Heimdall**, a cloud-native control-plane resource manager responsible for provisioning and tracking Linux/Windows VM fleets, disks, and containerized workloads.
- Developed services in **C++, Go and Python**, using **gRPC and REST APIs** for communication with dependent services and internal tooling.
- Designed resource-state models using **SQL and NoSQL databases**, selecting storage approaches based on consistency, query, and workload requirements.
- Worked with **Kubernetes-orchestrated containerized services** alongside traditional VM-based workloads.
- Owned features end-to-end, including architecture/design discussions, requirements analysis, implementation, code reviews, testing, and production support.
- Developed automated unit tests using **GoogleTest/GoogleMock** and integrated test execution through Ginkgo and TestRail.
- Investigated and resolved **release-blocking and customer-impacting production issues**, performing root-cause analysis and delivering fixes.
- Maintained build and deployment image tooling and generated **SBOMs** for deployment artifacts to support vulnerability tracking and compliance.
- Collaborated with dependent engineering teams to define interfaces, resource-management behavior, and service requirements.

Veritas Technologies — Principal Software Engineer, Shared Licensing Component (SLIC)

Jan 2023 – Dec 2024 | Pune, India

- Developed C++ licensing and entitlement functionality for **SLIC**, a shared license-validation component used across Veritas products.
- Designed and implemented features on Windows and ported/validated them across **Red Hat, OpenSUSE, AIX and Solaris**.
- Managed platform-specific builds of **OpenSSL and Poco** dependencies across supported operating systems.
- Investigated and remediated security vulnerabilities reported through **Black Duck**, assessing CVE impact and prioritizing fixes.
- Delivered updated SLIC SDKs and build artifacts to downstream product teams.
- Maintained artifacts in **JFrog Artifactory** and generated SBOMs to support downstream security and compliance requirements.
- Worked on cross-platform C++ development, debugging, build infrastructure, and release support.

Siemens Mobility — Senior Software Engineer / Technical Lead, Radio Block Center

Mar 2019 – Jan 2023 | Pune, India

- Led a **team of six engineers** developing safety-critical C/C++ software for a railway Radio Block Center responsible for processing train-position information and issuing movement authority.
- Delivered features across the complete development lifecycle, from requirements and design through implementation, static analysis, unit testing, and release.
- Conducted technical design discussions and **code reviews**, mentoring engineers and maintaining software quality.
- Investigated field issues using logs and diagnostics, performed root-cause analysis, and delivered corrective fixes.
- Developed Python-based internal automation tools for software/environment setup and unit-test baseline management, reducing repetitive manual activities across the team.
- Worked within a safety-critical development process involving requirements traceability, verification, and controlled releases.
- Owned monthly billing releases and forecast updates for the engagement.

Tata Elxsi Ltd. — Specialist, Automotive Embedded Systems

Jun 2015 – Mar 2019 | Pune, India

Automotive Infotainment — Panasonic Automotive

- Developed C/C++ components for smartphone-to-infotainment connectivity systems running on **QNX and Windows**.
- Implemented functionality for voice-command handling, touch/UI input, and embedded navigation.
- Integrated software with vehicle systems using **CAN bus** communication.
- Authored design documentation and performed production debugging using logs and core dumps.
- Diagnosed and resolved software defects across embedded platforms.

GM Instrument Cluster — Visteon

- Developed C/C++ software processing **CAN bus vehicle data** and driving instrument-cluster UI components for tire pressure, fuel level, speed, and tachometer information.
- Delivered features from system design through implementation, unit testing, and static analysis.
- Used GoogleTest/GoogleMock for automated testing.
- Applied design patterns to maintain consistent access to shared hardware interfaces.

Infosys Ltd. — Senior Software Engineer, Graphics Driver Development

Aug 2014 – Jun 2015 | Pune, India

- Worked on **AMD Radeon GPU driver and graphics software** for WinCE-based platforms.
- Ported the Linux **OpenGL/LLVM graphics call stack** to WinCE.
- Worked with the GPU driver's **DRM subsystem** and hardware-accelerated rendering pipeline.
- Diagnosed and resolved graphics-driver defects affecting rendering correctness and performance.
- Developed OpenGL test applications to validate rendering behavior across the graphics stack.
- Gained hands-on experience across GPU driver, graphics API, and hardware-acceleration layers.

Rockwell Collins India Pvt. Ltd. — Software Engineer, Onboard Maintenance System

Aug 2011 – Aug 2014 | Pune, India

- Developed **DO-178B compliant C/C++ software** for an avionics Onboard Maintenance System running on LynxOS.
- Developed applications for diagnostic reporting and display management.
- Implemented communication with aircraft Line Replaceable Units using **ARINC 429, ARINC 664/AFDX Ethernet, and XML-RPC**.
- Performed hardware-level debugging using **Trace32/Lauterbach**.
- Worked within a 20-person engineering program with requirements managed through IBM Rational DOORS and verification evidence aligned with DO-178B objectives.
- Developed and executed functional/unit tests and resolved integration defects.

TECHNICAL LEADERSHIP

- Led and mentored a **6-member engineering team** at Siemens Mobility.
- Conducted code reviews and design discussions across multiple projects.
- Owned features from requirements and architecture through implementation, testing, release, and production support.
- Experienced in cross-functional collaboration with QA, dependent engineering teams, product stakeholders, and customer-facing teams.
- Strong hands-on debugging and root-cause-analysis experience across application, embedded, and systems software.

EDUCATION

B.Tech — Uttar Pradesh Technical University (UPTU)
2010

Diploma in System Software — Pune University
2011